Nazmul Huda Badhon

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LinkedIn | Google Scholar | Portfolio

Graduate with peer-reviewed publications in deep learning and machine learning, with experience in convolutional neural networks. International academic exposure through a funded exchange study.

EDUCATION

Bachelor of Science — **Computer Science and Engineering** — Spring 2022–Fall 2025
Daffodil International University — Dhaka, Bangladesh
CGPA: 3.88 / 4.00

Computer Engineering (**Exchange Semester**) — Fall 2023
Dongseo University, GPA: 3.94 / 4.50 — Busan, South Korea

RESEARCH INTERESTS

- Designing computationally efficient deep neural network architectures.
- Developing robust and generalizable deep learning models for real-world applications.

RESEARCH PUBLICATIONS

Research Articles

1. **Nazmul Huda Badhon**, Imrus Salehin, Md. Tomal Ahmed Sajib, Md. Sakibul Hassan Rifat, S M Noman, Nazmun Nessa Moon. TLCNN: tabular data-based lightweight convolutional neural network for electricity energy demand prediction. Global Energy Interconnection (Q1), Elsevier, 14 Nov 2025. ISSN: 2590-0358.

2. Md. Tomal Ahmed Sajib, **Nazmul Huda Badhon**, Imrus Salehin, Md. Sakibul Hassan Rifat, Faysal Ahmmed, Nazmun Nessa Moon. A comparative deep learning methodology for plant insect image classification: assessment of CNN architectures and augmentation techniques. MethodsX (Q2), 103680, Elsevier, 17 Oct 2025. ISSN: 2215-0161.

Systematic Review

3. Imrus Salehin, Md. Tomal Ahmed Sajib, **Nazmul Huda Badhon**, Md. Sakibul Hassan Rifat, Nazrul Amin, Nazmun Nessa Moon. Systematic Literature Review of LLM-Large Language Model in Medical: Digital Health, Technology and Applications. Engineering Reports (Q2), 7.9: e70365, Wiley, 12 Sep 2025. ISSN: 2577-8196.

Data Article

4. Imrus Salehin, Mahbubur Rahman Khan, Ummya Habiba, **Nazmul Huda Badhon**, Nazmun Nessa Moon. BAU-Insectv2: An Agricultural Plant Insect Dataset for Deep Learning and Biomedical Image Analysis. Data in Brief (Q3), 110083, Elsevier, 23 Jan 2024. ISSN: 2352-3409.

Book Chapter

5. Imrus Salehin, **Nazmul Huda Badhon**, Md. Tomal Ahmed Sajib, Md. Sakibul Hassan Rifat, Nazmun Nessa Moon. Predicting Peak Energy Demand Using Machine Learning Techniques

for Efficient Electricity Supply Management. *Engineering Applications of AI for Demand Forecasting*, Taylor & Francis. Accepted on 24 Apr 2025 (In press).

PROJECTS

Selected Research Projects

Breast Cancer Classification using Hybrid Segmentation Deep Learning
- Built a two-stage U-Net and CNN-based hybrid model on the BUSI datasets.
- Achieved strong performance on accuracy and ROC metrics.

Lung and Colon Cancer Histopathology Classification
- Applied preprocessing on histopathology and evaluated multiple transfer learning models.
- Identified the best-performing model by accuracy and F1-score.

Vision Transformer–Based Medical Image Classification
- Designed preprocessing workflows for medical images and classification with ViT.
- Achieved 87% testing accuracy and improved transformer performance.

Additional Projects

Lost and Found Web Application

- Developed a full-stack web app with HTML, CSS, and JS for reporting and searching lost and found items.
- Implemented PHP-based backend, REST APIs, MySQL database, and secure user/admin authentication with image upload support.

SCHOLARSHIPS AND AWARDS

Global Korea Scholarship (GKS)–fully funded exchange scholarship — Fall 2023
Govt. of the Republic of Korea

Special Scholarship for Consistent Academic Excellence — Spring 2022–Fall 2025
Daffodil International University

Research Award for Outstanding Undergraduate Research Contribution — 2024
Division of Research, Daffodil International University

TECHNICAL SKILLS

- **Machine Learning & Deep Learning:** Convolutional Neural Networks, U-Net, Vision Transformers, Transfer Learning
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV
- **Programming:** Python, Java, C
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn
- **Tools & Platforms:** GitHub, Google Colab, LaTeX

ACADEMIC SERVICE & LEADERSHIP

Peer Reviewer Recognition — 2025
Journal: Digital Health Publisher: SAGE Publishing

Volunteer Internship 2024
International Affairs, Daffodil International University

Team Leader 2023
International Camping, Dongseo University, Busan, South Korea

REFERENCES

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